

Education Major GPA: 4.9/5.0 GPA: 4.6/5.0

Massachusetts Institute of Technology

Cambridge, MA

Fall 2014—Spring 2018

- Bachelor of Science in **Materials Science and Engineering** and Concentrations in **Mechanical Engineering and Economics**
- Relevant Coursework: Additive Manufacturing, Physical Metallurgy, Ceramics, Polymer Engineering, Mechanics of Materials, Structural Materials, Thermo, Material Structure, Design and Manufacturing II, How to Make (Almost) Anything

Experience

SpaceX – Lead Starlink Engineer

Materials Engineer 1

July 2018 - Aug 2020

Materials Engineer 2

Aug 2020 - Mar 2022

Senior Materials Engineer

Mar 2022 - Oct. 2022

Lead, Starlink Process Engineering (PCB Fab)

Oct 2022 - July 2024

Manager, Starlink Process Engineering (PCB Fab)

July 2024 - Jan. 2026

Manager, Starlink Process Engineering (Solar Cell Fab)

Jan. 2026 - Present

- Led a \$220M Capex project for insourcing printed circuit board production for SpaceX. Initially came up with idea of insourcing this critical piece part, and followed through with building a team up, learning all of the critical process details from scratch, designed the production line, presented the project to senior leadership for approval (Gwynne, Elon), performed acceptance testing of machines, developed my own highly detailed cost model, designed facility layouts, and succeeded in making a functional in-house PCB off of the line. Working to ramp up production now.
- Other projects I've been instrumental on over the last 6 years in Starlink involve selecting and testing hydrophobic coatings for the V1 radome, as well as evaluate alternative PSAs for the antenna assembly, finding/testing/formulating the soft epoxy resins myself that we ended up using throughout V1 and V2 antenna builds, along with developing numerous processes used throughout from scratch (stencil printing, flame treaters, oxide coatings for patch layers, balancing laminate, TIM selection/ processes, plastic selection for all antenna/enclosure parts, ETFE/epoxy radome structure (my ETFE film still in use today), various fun debacles, etc.)
- On V3, in addition to initial design and manufacturing bringup, I have led numerous cost cutting initiatives, including replacing dk spacers with a cheaper extruded plastic, owned switchover of patch layers to conductive printed ink, owned plastic selection / testing, worked to reduce ETFE sticker cost by qualifying cheaper PSAs and release films, researched and presented on insourcing stamped metal parts (which we did), etc, saving the company tens of millions.
- Currently managing a process engineering team that is insourcing solar cell production (at extreme volume) for Starlink satellites and Starlink AI satellites.

Apple – iPhone Product Design Engineer

Summer Intern

Summer 2017

- Researched and developed initial wireless charging technology for Apple Watch, AirPods, and AirPower Charging Mat, including simulation in Ansys electromagnetic software, modeling and fabricating initial mockups, and running testing
- Worked with external suppliers to help bring charging coil production up to rate, and worked through several DFM rounds, optimizing the part design
- Designed and fabricated several test setups through collaboration with electrical engineering, materials, and product design teams for benchmarking technologies and conducting user experience studies
- Extensive use of NX CAD, ANSYS mechanical, thermal, and electromagnetic simulation software

CEiiA (Portuguese Aeronautics/Mobility Company) – Engineering Intern

Summer Intern

Summer 2016

- Performed market analysis and research on battery management system (BMS) options for the UAS30 – a 30kg UAV
- Designed (in Eagle), purchased, and built a BMS PCB for the UAV power system
- Performed several composite (carbon fiber) layups for another intern's 3-axis gimbal design

UROP in the MIT Electrochemical Energy Lab

Undergraduate Researcher

Summer 2015 – Spring 2016

- Conducted research into catalysis and synthesized a series of doped nanomaterials that had high surface area and high

- conductivity for battery and fuel cell applications
- Characterized the oxides using XRD, BET, SEM, and three-electrode techniques

UROP in the MIT Grossman Group

Undergraduate Researcher

Winter 2014 – Spring 2016

- Designed and built a novel tool to determine the thermal conductivity of concrete disks with various additives in them (graphene, coal, graphite, carbon black, etc.)

Self-Employed

The Electrode Supply

Winter 2012 – Present

- Sell a variety of products (electrodes, chemicals, etc.) through eBay and on my website (<http://electrodesupply.com>)
- Sold for professional research (have sold to US National Labs, Universities, businesses) as well as amateur experimentation

Leadership and Activities

MIT Electric FSAE Team

Composites/Aero Subteam and Battery/Powertrain Subteam

Fall 2016 – Summer 2017

- Worked on designing, making, and testing (normal/shear strain, strength, etc.) composite layups for fairings/aero

MIT Electric Vehicle Team

Electric Racing Motorcycle Subteam

Spring 2015 – Fall 2017

- Worked on a small team to develop an electric racing motorcycle for the Pikes Peak International Hill Climb. We took second place in the electric motorcycle class (Race Video --- http://youtu.be/SqOuaNS0o_c)
- Responsibilities included integrating the sensors into the bike, logging telem. data, and working with the CAN Bus

Boy Scouts of America

Eagle Scout and Group Leader (SPL for 60+ teenagers)

2004 – 2014

- Active member of Scouting for 11 years – received Eagle Scout award May 2012

Skills/Tools

Tools: Characterization – Mechanical Testing (Instron), Raman Spectroscopy, BET, XRD, SEM, DSC, STM, Optical Microscopy
Advanced Shop Experience and Electronics Knowledge – CNC Mill, Lathe, Laser Cutter, Waterjet, 3D Printing, PCB Fab

Computer: NX, ANSYS, SolidWorks, MasterCam, Altium, Matlab, Mathematica, Python, Minitab

Projects: Built a fully functioning STM (capable of ‘seeing’ atoms), Designed and built custom injection molded yo-yo’s, etc.

Certifications: Hazwopper trained (tech level)

About Me: Avid backpacker, camper, hiker, rock climber, canoer, rower, cooker, baker, entrepreneur, and photographer!

Publications/Patents

- [Screening Oxide Support Materials for OER Catalysts in Acid](#)
 - Binghong Han, Marcel Risch, **Samuel Belden**, Seonggyu Lee, Domnik Bayer, Eva Mutoro and Yang Shao-Horn; Published 20 July 2018, Journal of The Electrochemical Society
- [US11652286B2: Antenna apparatus having adhesive coupling](#)
 - David Milroy, **Samuel Belden**, Published 2020, Filed by Space Exploration Technology Corp
- [US20230142216A1: Radome assembly having an outer layer](#)
 - Darya Malkova, David Milroy, **Samuel Belden**, Published 2022, Filed by Space Exploration Technology Corp